KAIZEN MATHS

Lesson starter

Advanced Factorisation: Grouping, Difference of Squares and Cubics

Basic Common Factors

Name _____

S1

$12a + 9$

S2

$x^4 - 2x^3$

S3

$25r^3 - 30r^2$

S4

$8a - 20$

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Exit ticket

Advanced Factorisation: Grouping, Difference of Squares and Cubics

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Name _____

E1

$y^3 + 2y^2$

E2

$42r^4 - 36r^3$

E3

$16a - 28$

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Teacher answers & working

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S1

$12a + 9$

Answer:

$3(4a + 3)$

Working:

Method focus: find the largest number that divides every coefficient.

Step 1: Compare the coefficients 12 and 9.

Step 2: The highest common numerical factor is 3.

$12a + 9 = 3(4a + 3)$

Step 3: Check by multiplying 3 back into the bracket.

S2

$x^4 - 2x^3$

Answer:

$x^3(x - 2)$

Working:

Method focus: take out the smallest shared power of the variable.

Step 1: Compare the powers of x: x^4 and x^3.

Step 2: The smallest shared power is x^3.

$x^4 - 2x^3 = x^3(x - 2)$

Step 3: Check by multiplying x^3 through the bracket.

S3

$25r^3 - 30r^2$

Answer:

$5r^2(5r - 6)$

Working:

Method focus: take out both the numerical HCF and the shared variable factor.

Step 1: Find the numerical HCF: HCF(25, 30) = 5.

Step 2: Find the shared variable factor: r^2.

Step 3: Combine them to get the full HCF $5r^2$.

$25r^3 - 30r^2 = 5r^2(5r - 6)$

Step 4: The terms inside the bracket now have no common factor left.

Teacher answers & working

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S4

$8a - 20$

Answer:

$4(2a - 5)$

Working:

Method focus: find the largest number that divides every coefficient.

Step 1: Compare the coefficients 8 and 20.

Step 2: The highest common numerical factor is 4.

$8a - 20 = 4(2a - 5)$

Step 3: Check by multiplying 4 back into the bracket.

E1

$y^3 + 2y^2$

Answer:

$y^2(y + 2)$

Working:

Method focus: take out the smallest shared power of the variable.

Step 1: Compare the powers of y: y^3 and y^2.

Step 2: The smallest shared power is y^2.

$y^3 + 2y^2 = y^2(y + 2)$

Step 3: Check by multiplying y^2 through the bracket.

E2

$42r^4 - 36r^3$

Answer:

$6r^3(7r - 6)$

Working:

Method focus: take out both the numerical HCF and the shared variable factor.

Step 1: Find the numerical HCF: HCF(42, 36) = 6.

Step 2: Find the shared variable factor: r^3.

Step 3: Combine them to get the full HCF $6r^3$.

$42r^4 - 36r^3 = 6r^3(7r - 6)$

Step 4: The terms inside the bracket now have no common factor left.

Teacher answers & working

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E3

16a - 28

Answer:

4(4a - 7)

Working:

Method focus: find the largest number that divides every coefficient.

Step 1: Compare the coefficients 16 and 28.

Step 2: The highest common numerical factor is 4.

16a - 28 = 4(4a - 7)

Step 3: Check by multiplying 4 back into the bracket.